A high-temperature CLT for the SYK free energy: a two-replica cluster-expansion reduction

Abstract

This note gives a detailed Gaussian-chaos proof of the finite-dimensional high-temperature central limit theorem for the Gaussian SYK free energy, conditional on one explicitly isolated two-marked susceptibility estimate. The analytic derivative part of the LLN is proved from locally uniform high-temperature convergence by a deterministic convexity argument. The first chaos vanishes by Majorana sign-gauge symmetry. The second chaos is computed exactly and is a scalar multiple of $\sum_I (g_I^2 - 1)$. The fixed finite higher chaoses are proved to be negligible. The infinite chaos tail is no longer controlled by a uniform all-orders marked-core estimate; instead it is controlled by the Hilbert–Schmidt norm of the Hessian, via the standard Malliavin identity. The proof of the two-marked high-temperature susceptibility is pushed as far as possible. The Duhamel cumulant identities, the dependency-graph tree bound for the zero-source Duhamel coefficients, and the connected parity counting lemma are proved explicitly. The remaining input is isolated as a resummed two-replica L^2 cluster-expansion estimate. This assumption is deliberately stated separately from the proved coefficientwise and counting lemmas, because repeated hyperedges make the naive occurrence-level tree summation too crude.

Status. The result of Gamarnik–Pernice–Schmidhuber–Zlokapa gives the high-temperature free-energy limit by a sparse-hypergraph and saddle-point analysis, while Anschuetz–Chen–Kiani–King give an annealed/quenched comparison strong enough for the LLN level. The proof below uses these only as the locally uniform LLN input. The CLT-level reduction is the Gaussian-chaos analysis. The remaining input is reduced to a two-point, rather than all-orders, marked estimate: a uniform L^2 bound on the Kubo covariance matrix of two SYK monomials. This is stated in Assumption 1.3.

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Reader's guide and dependency of assumptions

This note is meant to separate the purely Gaussian-chaos part of the CLT from one genuinely high-temperature cluster-expansion input. The only input which is not proved in the present note is the resummed two-replica L^2 cluster expansion, Assumption 8.17. Once this assumption is supplied, the rest of the CLT proof is self-contained.

The logical dependencies are as follows. The LLN input is Assumption 1.1. The remaining CLT-level input is first stated in the softer form of the two-marked susceptibility estimate, Assumption 1.3. Section 8 then reduces this susceptibility estimate to the more structural cluster assumption, Assumption 8.17. In symbols,

$$\begin{aligned}&\text{Assumption 1.1} + \text{Assumption 8.17}\\ \Longrightarrow\ &\text{Assumption 1.3} \Longrightarrow \text{Theorem 1.4}.\end{aligned}$$

Equivalently, after the LLN input, the only genuinely unproved input is Assumption 8.17.

More explicitly, the proof is organized through the following chain.

$$\begin{aligned}&\text{resummed two-replica } L^2 \text{ cluster expansion}\\ \Longrightarrow\ &\text{two-marked susceptibility} \Longrightarrow \text{Hilbert–Schmidt Hessian bound}\\ \Longrightarrow\ &\text{infinite chaos tail closure} \Longrightarrow \text{rank-one second-chaos CLT}.\end{aligned}$$

The parts proved in this note are the following.

(i) The derivative convergence $N^{-1}\partial_\beta \mathbb{E} L_N(\beta) \to \ell_q'(\beta)$ is derived from the locally uniform LLN by convexity; see Proposition 2.2.

(ii) The first chaos vanishes by Majorana sign-gauge symmetry, and the second chaos is computed exactly; see Lemma 3.3, Lemma 4.2, and Proposition 4.3.

(iii) The second chaos gives the whole limiting Gaussian field once the higher chaoses are negligible; see Lemma 5.2.

(iv) The two-marked susceptibility implies a Hilbert–Schmidt bound on the Hessian, and this yields the infinite chaos tail closure by the Malliavin identity; see Lemma 6.1, Lemma 6.2, and Proposition 6.3.

(v) The fixed finite chaoses are negligible by a finite marked-diagram counting estimate; see Proposition 7.4.

(vi) The zero-source Duhamel cumulants admit a dependency-graph tree bound, and the connected parity counting lemma is proved; see Proposition 8.8 and Lemma 8.11.

The parts not proved in this note are deliberately isolated. The naive single-replica activity expansion fails because the absolute activity has size $N^{(q-1)/2}$ near a marked edge. The naive coefficientwise tree expansion also fails because repeated hyperedges create occurrence-level entropy which should be resummed. These failures are recorded in Subsection 8.5. Thus the correct missing input is not a coefficientwise tree estimate, but the resummed two-replica estimate of Assumption 8.17.

For a first reading, one can read Sections 1–7 for the CLT reduction, then read the statement of Assumption 8.17 and Proposition 8.19. The remaining part of Section 8 explains why this assumption is the natural high-temperature input and why the more naive alternatives are too strong.

1 Model and main theorem

Fix an even integer $q \geq 4$. Let N be even and let $\psi_1, \dots, \psi_N$ be Majorana operators on $\mathbb{C}^{2^{N/2}}$ satisfying

$$\psi_i \psi_j + \psi_j \psi_i = 2\delta_{ij} \mathbb{I}.$$

Thus $\psi_i^2 = \mathbb{I}$. For $I = \{i_1 < \cdots < i_q\} \subset [N]$, set

$$\psi_I = i^{q/2} \psi_{i_1} \cdots \psi_{i_q}.$$

This is the paper notation for the Lean term $(\text{phase}\, q) \cdot M.\, \text{psiProduct}(I)$. Then ψ_I is Hermitian and $\psi_I^2 = \mathbb{I}$.

Let

$$\mathcal{I}_N = \{I \subset [N] : |I| = q\}, \qquad M_N = |\mathcal{I}_N| = \binom{N}{q}.$$

Thus $\mathcal{I}_N$ is the paper notation for Lean's InteractionIndex $N\, q$. Let $(g_I)_{I \in \mathcal{I}_N}$ be i.i.d. standard Gaussian random variables and set

$$\sigma_N^2 = \frac{(q-1)!}{N^{q-1}}.$$

The Gaussian SYK Hamiltonian is

$$H_N(g) = \sigma_N \sum_{I \in \mathcal{I}_N} g_I \psi_I.$$

Equivalently, this is the Lean Hamiltonian with coupling $J_I = \sigma_N g_I$. We use normalized trace $\mathrm{tr} = 2^{-N/2}\,\mathrm{Tr}$ and define

$$Z_N(\beta) = \mathrm{tr} \exp(-\beta H_N(g)), \qquad L_N(\beta) = \log Z_N(\beta).$$

For an observable A, write

$$\langle A \rangle_{\beta,g} = \frac{\mathrm{tr}(A e^{-\beta H_N(g)})}{\mathrm{tr}(e^{-\beta H_N(g)})}.$$

For $I, J \in \mathcal{I}_N$, define the Kubo connected covariance

$$\mathcal{C}_{I,J}(\beta, g) = \int_0^1 \left\{ \frac{\mathrm{tr}\left(\psi_I e^{-s\beta H_N(g)} \psi_J e^{-(1-s)\beta H_N(g)}\right)}{Z_N(\beta)} - \langle \psi_I \rangle_{\beta,g} \langle \psi_J \rangle_{\beta,g} \right\} ds. \tag{1}$$

The rescaled centered fluctuation is

$$X_N(\beta) = N^{q/2-1}\big(L_N(\beta) - \mathbb{E} L_N(\beta)\big).$$

Let P_r be the orthogonal projection in L^2 of the Gaussian disorder onto the r-th homogeneous Wiener chaos.

The proof uses the following LLN input from the high-temperature free-energy analysis. The derivative part is not assumed separately; it is proved in Proposition 2.2.

Assumption 1.1 (Locally uniform high-temperature LLN)**.** *There exist $\beta_0 = \beta_0(q) > 0$ and an analytic function $\ell_q : (-\beta_0, \beta_0) \to \mathbb{R}$ such that, for every compact $K \subset (-\beta_0, \beta_0)$,*

$$\sup_{\beta \in K} \left| \frac{1}{N} \mathbb{E} L_N(\beta) - \ell_q(\beta) \right| \to 0.$$

Remark 1.2. *In the present normalization, Assumption 1.1 is the precise LLN form extracted from the high-temperature result of [GPSZ26], together with the Gaussian universality used there and the annealed/quenched comparison of [ACKK25]. The latter gives an error $O_q(\beta^2 N^{-q/2})$ at the level of free-energy density, so it is smaller than the LLN scale. The proof below does not use any CLT statement from these references.*

The CLT-level input is the following high-temperature two-marked susceptibility estimate. Section 8 reduces this estimate to a single explicit, plausible, but presently unproved input: the resummed two-replica L^2 cluster-expansion estimate in Assumption 8.17.

Assumption 1.3 (Two-marked high-temperature susceptibility)**.** *There exists $\beta_{\mathrm{sus}} = \beta_{\mathrm{sus}}(q) > 0$ such that, for every compact $K \subset (-\beta_{\mathrm{sus}}, \beta_{\mathrm{sus}})$, the quantity*

$$S_{q,K} := \sup_{N\geq 1} \sup_{\beta\in K} \sup_{I\in\mathcal{I}_N} \sum_{J\in\mathcal{I}_N} \mathbb{E}\left[|\mathcal{C}_{I,J}(\beta, g)|^2\right] < \infty. \tag{2}$$

Theorem 1.4 (High-temperature CLT from two-marked susceptibility)**.** *Assume Assumptions 1.1 and 1.3. Let*

$$\beta_* < \min\{\beta_0, \beta_{\mathrm{sus}}\}.$$

Then, for every finite set $\beta_1, \dots, \beta_k \in (-\beta_*, \beta_*)$,

$$(X_N(\beta_1), \dots, X_N(\beta_k)) \Rightarrow \left(\sqrt{\frac{q!}{2}}\,\beta_1 \ell_q'(\beta_1)\xi, \dots, \sqrt{\frac{q!}{2}}\,\beta_k \ell_q'(\beta_k)\xi\right),$$

where $\xi \sim \mathcal{N}(0,1)$. Equivalently, the limiting covariance is

$$\mathrm{Cov}(\mathcal{G}_q(\beta), \mathcal{G}_q(\beta')) = \frac{q!}{2}\beta\ell_q'(\beta)\beta'\ell_q'(\beta').$$

2 Derivative LLN from convexity

Lemma 2.1 (Convex convergence implies derivative convergence)**.** *Let $I \subset \mathbb{R}$ be an open interval. Let $f_N : I \to \mathbb{R}$ be differentiable convex functions and assume that $f_N \to f$ locally uniformly on I, where $f \in C^1(I)$. Then, for every compact $K \subset I$,*

$$\sup_{x\in K} |f_N'(x) - f'(x)| \to 0.$$

Proof. Fix a compact $K \subset I$. Choose $h_0 > 0$ such that $K + [-h_0, h_0] \subset I$. For $0 < h < h_0$ and $x \in K$, convexity gives

$$\frac{f_N(x) - f_N(x-h)}{h} \leq f_N'(x) \leq \frac{f_N(x+h) - f_N(x)}{h}. \tag{3}$$

Put

$$\epsilon_N(h) = \sup_{y\in K+[-h,h]} |f_N(y) - f(y)|.$$

Then $\epsilon_N(h) \to 0$ for fixed h. From (3),

$$f_N'(x) - f'(x) \leq \frac{f(x+h) - f(x)}{h} - f'(x) + \frac{2\epsilon_N(h)}{h},$$

and

$$f_N'(x) - f'(x) \geq \frac{f(x) - f(x-h)}{h} - f'(x) - \frac{2\epsilon_N(h)}{h}.$$

Since f' is uniformly continuous on $K + [-h_0, h_0]$, the deterministic difference-quotient errors converge to zero uniformly on K as $h \downarrow 0$. First choose h and then choose N to make $\epsilon_N(h)/h$ small. This proves the claim. □

Proposition 2.2 (Derivative-strength LLN)**.** *Under Assumption 1.1, for every compact $K \subset (-\beta_0, \beta_0)$,*

$$\sup_{\beta\in K} \left|\frac{1}{N}\partial_\beta \mathbb{E} L_N(\beta) - \ell_q'(\beta)\right| \to 0. \tag{4}$$

Proof. Let $f_N(\beta) = N^{-1}\mathbb{E}L_N(\beta)$. For fixed disorder, $\beta \mapsto L_N(\beta)$ is convex, because $L_N''(\beta)$ is the thermal variance of $H_N(g)$. Therefore f_N is convex. Differentiability follows by differentiating under the Gaussian expectation; this is justified by $\|H_N(g)\| \le \sigma_N \sum_{I\in\mathcal{I}_N} |g_I|$ and Gaussian exponential integrability in finite dimension. Assumption 1.1 gives locally uniform convergence of f_N to the analytic, hence C^1, function ℓ_q. Lemma 2.1 gives (4). □

3 Hermite decomposition and gauge selection

Let γ_N be the product Gaussian law of $(g_I)_{I\in\mathcal{I}_N}$. For a multi-index $\alpha = (\alpha_I)_{I\in\mathcal{I}_N}$ define

$$|\alpha| = \sum_{I\in\mathcal{I}_N} \alpha_I, \qquad \alpha! = \prod_{I\in\mathcal{I}_N} \alpha_I!, \qquad H_\alpha(g) = \prod_{I\in\mathcal{I}_N} h_{\alpha_I}(g_I),$$

where h_k is the probabilists' Hermite polynomial.

Lemma 3.1 (Hermite coefficient formula)**.** *For every fixed $\beta \in \mathbb{R}$,*

$$L_N(\beta) - \mathbb{E}L_N(\beta) = \sum_{r\ge 1} P_r L_N(\beta)$$

in $L^2(\gamma_N)$, and

$$P_r L_N(\beta) = \sum_{|\alpha|=r} c_{N,\alpha}(\beta) H_\alpha(g), \qquad c_{N,\alpha}(\beta) = \frac{1}{\alpha!}\mathbb{E}[\partial^\alpha L_N(\beta)]. \tag{5}$$

Proof. The map $g \mapsto L_N(\beta)$ is C^∞ and has at most exponential growth in the finite-dimensional Gaussian vector g. Hence its Hermite expansion converges in $L^2(\gamma_N)$. Orthogonality gives

$$P_r L_N(\beta) = \sum_{|\alpha|=r} \frac{\mathbb{E}[L_N(\beta)H_\alpha(g)]}{\alpha!} H_\alpha(g).$$

Repeated Gaussian integration by parts gives

$$\mathbb{E}[L_N(\beta)H_\alpha(g)] = \mathbb{E}[\partial^\alpha L_N(\beta)],$$

proving (5). □

For $s = (s_1, \dots, s_N) \in \{\pm 1\}^N$ define

$$\chi_s(I) = \prod_{a\in I} s_a, \qquad (\chi_s g)_I = \chi_s(I) g_I.$$

Lemma 3.2 (Majorana sign-gauge invariance)**.** *For every $s \in \{\pm 1\}^N$ and every g,*

$$L_N(\beta, g) = L_N(\beta, \chi_s g).$$

Proof. The family $(s_a\psi_a)_{a=1}^N$ satisfies the same Clifford relations as $(\psi_a)_{a=1}^N$. By uniqueness of the irreducible complex Clifford representation, there exists a unitary U_s such that

$$U_s \psi_a U_s^* = s_a \psi_a \qquad \text{for all } a \in [N].$$

Hence $U_s \psi_I U_s^* = \chi_s(I)\psi_I$ and

$$U_s H_N(g) U_s^* = H_N(\chi_s g).$$

The normalized trace is unitary invariant, so $Z_N(\beta, g) = Z_N(\beta, \chi_s g)$. Taking logarithms proves the claim. □

Lemma 3.3 (Gauge parity selection rule)**.** *Let α be a multi-index. If there exists $a \in [N]$ such that*

$$\sum_{I \ni a} \alpha_I \equiv 1 \pmod 2, \tag{6}$$

then $c_{N,\alpha}(\beta) = 0$. In particular,

$$P_1 L_N(\beta) = 0,$$

and for $I \neq J$,

$$\mathbb{E}[\partial_{g_I} \partial_{g_J} L_N(\beta)] = 0.$$

Proof. Since $\chi_s g$ has the same law as g and $L_N(\beta, \chi_s g) = L_N(\beta, g)$,

$$c_{N,\alpha}(\beta) = \frac{1}{\alpha!} \mathbb{E}[L_N(\beta, g) H_\alpha(\chi_s g)].$$

Since $h_k(-x) = (-1)^k h_k(x)$,

$$H_\alpha(\chi_s g) = \prod_{a=1}^{N} s_a^{\sum_{I \ni a} \alpha_I} H_\alpha(g).$$

If (6) holds, choose $s_a = -1$ and $s_b = 1$ for $b \neq a$. Then $c_{N,\alpha}(\beta) = -c_{N,\alpha}(\beta)$, so $c_{N,\alpha}(\beta) = 0$. The first-chaos statement follows by taking $\alpha = e_I$. If $I \neq J$, choose $a \in I \triangle J$ and take $\alpha = e_I + e_J$. □

4 Exact second chaos

Lemma 4.1 (First derivative identity)**.** *For every $I \in \mathcal{I}_N$,*

$$\partial_{g_I} L_N(\beta) = -\beta \sigma_N \langle \psi_I \rangle_{\beta, g}. \tag{7}$$

Proof. By the derivative formula for the matrix exponential,

$$\partial_{g_I} e^{-\beta H_N(g)} = -\beta \sigma_N \int_0^1 e^{-(1-t)\beta H_N(g)} \psi_I e^{-t\beta H_N(g)} \, dt.$$

After taking normalized trace, cyclicity makes the integrand independent of t and equal to $\operatorname{tr}(\psi_I e^{-\beta H_N(g)})$. Dividing by $Z_N(\beta)$ proves (7). □

Lemma 4.2 (Exact second chaos)**.** *Set*

$$a_N(\beta) = \frac{\beta \partial_\beta \mathbb{E} L_N(\beta)}{2M_N}. \tag{8}$$

Then

$$P_2 L_N(\beta) = a_N(\beta) \sum_{I \in \mathcal{I}_N} (g_I^2 - 1). \tag{9}$$

Moreover,

$$2M_N a_N(\beta) = \beta \partial_\beta \mathbb{E} L_N(\beta). \tag{10}$$

Proof. The second chaos has the form

$$P_2 L_N(\beta) = \sum_{I \in \mathcal{I}_N} c_{2e_I}(\beta)(g_I^2 - 1) + \sum_{I<J} c_{e_I + e_J}(\beta) g_I g_J.$$

By Lemma 3.3, the off-diagonal coefficients vanish. The diagonal coefficient is independent of I by permutation symmetry of the Majorana labels and exchangeability of the disorder. Thus

$$P_2 L_N(\beta) = \widetilde{a}_N(\beta) \sum_{I \in \mathcal{I}_N} (g_I^2 - 1)$$

for the common diagonal coefficient $\widetilde{a}_N(\beta)$.

By Lemma 3.1,

$$\widetilde{a}_N(\beta) = \frac{1}{2}\mathbb{E}[\partial_{g_I}^2 L_N(\beta)].$$

Summing over I and using Gaussian integration by parts,

$$2M_N\widetilde{a}_N(\beta) = \sum_{I\in\mathcal{I}_N} \mathbb{E}[\partial_{g_I}^2 L_N(\beta)] = \sum_{I\in\mathcal{I}_N} \mathbb{E}[g_I \partial_{g_I} L_N(\beta)].$$

By Lemma 4.1,

$$\sum_{I\in\mathcal{I}_N} g_I \partial_{g_I} L_N(\beta) = -\beta\sigma_N \sum_{I\in\mathcal{I}_N} g_I \langle \psi_I \rangle_{\beta,g} = \beta\partial_\beta L_N(\beta).$$

Taking expectation proves $\widetilde{a}_N(\beta) = \beta\partial_\beta \mathbb{E}L_N(\beta)/(2M_N)$, hence (8) and (10). □

Proposition 4.3 (Exact reduction to the diagonal second chaos)**.** *For every fixed $\beta \in \mathbb{R}$,*

$$L_N(\beta) - \mathbb{E}L_N(\beta) = \frac{\beta\partial_\beta \mathbb{E}L_N(\beta)}{2M_N} \sum_{I\in\mathcal{I}_N} (g_I^2 - 1) + R_N(\beta),$$

where

$$R_N(\beta) = \sum_{r\geq 3} P_r L_N(\beta).$$

Proof. Combine the Hermite decomposition, the identity $P_1 L_N = 0$, and Lemma 4.2. □

5 The second-chaos CLT

Lemma 5.1 (Asymptotic coefficient)**.** *Under Assumption 1.1, for every compact $K \subset (-\beta_0, \beta_0)$,*

$$\sup_{\beta\in K} \left| N^{q/2-1}\sqrt{2M_N}\, a_N(\beta) - \sqrt{\frac{q!}{2}}\, \beta \ell_q'(\beta) \right| \to 0.$$

Proof. By (10),

$$a_N(\beta) = \frac{\beta\partial_\beta \mathbb{E}L_N(\beta)}{2M_N}. \tag{11}$$

Proposition 2.2 gives, uniformly for $\beta \in K$,

$$\partial_\beta \mathbb{E}L_N(\beta) = N\ell_q'(\beta) + o(N).$$

Since $M_N = \binom{N}{q} \sim N^q/q!$,

$$\frac{N^{q/2}}{\sqrt{2M_N}} \to \sqrt{\frac{q!}{2}}.$$

Substituting these estimates into (11) proves the result. □

Lemma 5.2 (Second-chaos CLT)**.** *Under Assumption 1.1, for any fixed $\beta_1, \dots, \beta_k \in (-\beta_0, \beta_0)$,*

$$\left(N^{q/2-1}P_2 L_N(\beta_1), \dots, N^{q/2-1}P_2 L_N(\beta_k)\right) \Rightarrow \left(\sqrt{\frac{q!}{2}}\,\beta_1\ell_q'(\beta_1)\xi, \dots, \sqrt{\frac{q!}{2}}\,\beta_k\ell_q'(\beta_k)\xi\right),$$

where $\xi \sim \mathcal{N}(0,1)$.

Proof. Define

$$S_n = \frac{1}{\sqrt{2M_N}} \sum_{I \in \mathcal{I}_N} (g_I^2 - 1).$$

Since the variables $g_I^2 - 1$ are i.i.d., centered, have variance 2, and have finite third moment, the Lindeberg CLT gives

$$S_n \Rightarrow \xi, \qquad \xi \sim \mathcal{N}(0,1).$$

By (9),

$$N^{q/2-1} P_2 L_N(\beta) = \left(N^{q/2-1} \sqrt{2M_N}\, a_N(\beta) \right) S_n.$$

Lemma 5.1 identifies the deterministic coefficient. The joint convergence follows because all coordinates are deterministic multiples of the same scalar S_n. □

6 Higher-chaos closure from two-marked susceptibility

The next two elementary facts replace the former all-orders marked-tail assumption by a second-derivative estimate.

Lemma 6.1 (Kubo covariance identity)**.** *For every $I, J \in \mathcal{I}_N$ and every $\beta \in \mathbb{R}$,*

$$\partial_{g_I} \partial_{g_J} L_N(\beta) = \beta^2 \sigma_N^2 \mathcal{C}_{I,J}(\beta, g). \tag{12}$$

Consequently, under Assumption 1.3, for every compact $K \subset (-\beta_{\mathrm{sus}}, \beta_{\mathrm{sus}})$, with $\beta_K = \sup_{\beta \in K} |\beta|$,

$$\sup_{\beta \in K} \mathbb{E} \sum_{I,J \in \mathcal{I}_N} |\partial_{g_I} \partial_{g_J} L_N(\beta)|^2 \le \beta_K^4 \frac{((q-1)!)^2}{q!} S_{q,K} N^{2-q}.$$

Proof. Lemma 4.1 gives

$$\partial_{g_I} L_N(\beta) = -\beta \sigma_N \langle \psi_I \rangle_{\beta,g}.$$

Differentiate $\langle \psi_I \rangle_{\beta,g}$ with respect to g_J. Using the Duhamel formula

$$\partial_{g_J} e^{-\beta H_N(g)} = -\beta \sigma_N \int_0^1 e^{-s\beta H_N(g)} \psi_J e^{-(1-s)\beta H_N(g)}\, ds$$

and differentiating the denominator $Z_N(\beta)$ gives

$$\begin{aligned} \partial_{g_J} \langle \psi_I \rangle_{\beta,g} &= -\beta \sigma_N \int_0^1 \left\{ \frac{\operatorname{tr}\left(\psi_I e^{-s\beta H_N(g)} \psi_J e^{-(1-s)\beta H_N(g)} \right)}{Z_N(\beta)} - \langle \psi_I \rangle_{\beta,g} \langle \psi_J \rangle_{\beta,g} \right\} ds \\ &= -\beta \sigma_N \mathcal{C}_{I,J}(\beta, g). \end{aligned}$$

This proves (12). By Assumption 1.3,

$$\begin{aligned} \sup_{\beta \in K} \mathbb{E} \sum_{I,J \in \mathcal{I}_N} |\partial_{g_I} \partial_{g_J} L_N(\beta)|^2 &\le \beta_K^4 \sigma_N^4 \sum_{I \in \mathcal{I}_N} S_{q,K} \\ &\le \beta_K^4 S_{q,K} \sigma_N^4 M_N \le \beta_K^4 \frac{((q-1)!)^2}{q!} S_{q,K} N^{2-q}, \end{aligned}$$

because $M_N \le N^q/q!$ and $\sigma_N^4 = ((q-1)!)^2 N^{-2(q-1)}$. □

Lemma 6.2 (Second Malliavin derivative and chaos tail)**.** *Let $F \in \mathbb{D}^{2,2}(\gamma_N)$ have Hermite expansion $F = \sum_{r \ge 0} P_r F$. Then*

$$\mathbb{E} \left\| D^2 F \right\|_{\mathrm{HS}}^2 = \sum_{r \ge 2} r(r-1) \left\| P_r F \right\|_2^2. \tag{13}$$

In particular, for every $R \ge 2$,

$$\sum_{r > R} \left\| P_r F \right\|_2^2 \le \frac{1}{R(R-1)} \mathbb{E} \left\| D^2 F \right\|_{\mathrm{HS}}^2. \tag{14}$$

Proof. It is enough to check the identity for a finite Hermite polynomial and then pass to the limit by monotone convergence of the chaos-side and closability of the Malliavin derivative. For a multi-index α,

$$\partial_{g_I}\partial_{g_J}H_\alpha = \begin{cases} \alpha_I\alpha_J H_{\alpha-e_I-e_J}, & I \neq J, \\ \alpha_I(\alpha_I-1)H_{\alpha-2e_I}, & I = J. \end{cases}$$

Using $\mathbb{E}[H_\alpha^2] = \alpha!$ and summing over I, J, one gets, for $|\alpha| = r$,

$$\mathbb{E}\left\|D^2 H_\alpha\right\|_{\mathrm{HS}}^2 = r(r-1)\alpha!.$$

Orthogonality of Hermite polynomials of different multi-index then gives (13). Since $r(r-1) \geq R(R-1)$ for $r > R$, (14) follows. □

Proposition 6.3 (Two-marked susceptibility implies higher-chaos closure)**.** *Under Assumption 1.3, for every compact $K \subset (-\beta_{\mathrm{sus}}, \beta_{\mathrm{sus}})$,*

$$\sup_{\beta\in K}\mathbb{E}\left[\left|N^{q/2-1}\sum_{r\geq 3}P_r L_N(\beta)\right|^2\right] \to 0. \tag{15}$$

Proof. Orthogonality of homogeneous chaoses gives

$$\mathbb{E}\left[\left|N^{q/2-1}\sum_{r\geq 3}P_r L_N(\beta)\right|^2\right] = N^{q-2}\sum_{r\geq 3}\|P_r L_N(\beta)\|_2^2.$$

Fix $R \geq 3$. Split the sum into $3 \leq r \leq R$ and $r > R$. Proposition 7.4 gives

$$\sup_{\beta\in K} N^{q-2}\sum_{r=3}^{R}\|P_r L_N(\beta)\|_2^2 \to 0.$$

For the tail, apply Lemma 6.2 to $F = L_N(\beta) - \mathbb{E}L_N(\beta)$. Since $D^2F = D^2L_N(\beta)$, Lemma 6.1 gives, with $\beta_K = \sup_{\beta\in K}|\beta|$,

$$\sup_{\beta\in K} N^{q-2}\sum_{r>R}\|P_r L_N(\beta)\|_2^2 \leq \frac{\beta_K^4((q-1)!)^2 S_{q,K}}{q!R(R-1)}.$$

Taking $\limsup_{N\to\infty}$ and then $R \to \infty$ proves (15). □

Proof of Theorem 1.4. By Proposition 4.3,

$$X_N(\beta) = N^{q/2-1}P_2 L_N(\beta) + N^{q/2-1}R_N(\beta).$$

Lemma 5.2 gives the finite-dimensional convergence of the first term. Proposition 6.3 shows that the second term is negligible in L^2, uniformly on compact subintervals of $(-\beta_*, \beta_*)$. Slutsky's theorem proves the finite-dimensional CLT. □

7 Fixed finite chaos estimates

This section proves the fixed finite chaos estimate used in the Hessian-tail argument. Thus the only infinite-tail input needed for the CLT is the second-derivative estimate supplied by Assumption 1.3.

Lemma 7.1 (Log-partition derivative bound)**.** *Let K and $B_1, \dots, B_r$ be Hermitian matrices. Put*

$$\Phi(x_1, \dots, x_r) = \log \operatorname{tr} \exp\left(K + \sum_{j=1}^{r} x_j B_j\right).$$

Then

$$|\partial_{x_1} \cdots \partial_{x_r} \Phi(0)| \le 2^r r! \prod_{j=1}^{r} \|B_j\|. \tag{16}$$

Proof. Let $\mathcal{Z}(x) = \operatorname{tr} \exp(K + \sum_j x_j B_j)$ and $Z_0 = \mathcal{Z}(0)$. The Duhamel formula gives, for distinct variables $x_{i_1}, \dots, x_{i_m}$,

$$\partial_{i_1} \cdots \partial_{i_m} \mathcal{Z}(0) = \sum_{\pi \in S_m} \int_{\Delta_m} \operatorname{tr}\left(e^{t_0 K} B_{i_{\pi(1)}} e^{t_1 K} \cdots B_{i_{\pi(m)}} e^{t_m K}\right) dt. \tag{17}$$

Here $\Delta_m = \{t_j \ge 0 : \sum_{j=0}^{m} t_j = 1\}$. By noncommutative Holder, each trace in (17) is bounded in absolute value by

$$Z_0 \prod_{j=1}^{m} \|B_{i_j}\|.$$

Therefore

$$\left|\frac{\partial_{i_1} \cdots \partial_{i_m} \mathcal{Z}(0)}{Z_0}\right| \le m! \prod_{j=1}^{m} \|B_{i_j}\|. \tag{18}$$

The derivative of $\log \mathcal{Z}$ is obtained from the usual moment-cumulant formula over set partitions:

$$\partial_1 \cdots \partial_r \log \mathcal{Z}(0) = \sum_{\Pi \in \mathsf{P}([r])} (-1)^{|\Pi|-1} (|\Pi| - 1)! \prod_{A \in \Pi} \frac{\partial_A \mathcal{Z}(0)}{Z_0}.$$

Applying (18) to every block gives

$$|\partial_1 \cdots \partial_r \log \mathcal{Z}(0)| \le \left(\sum_{\Pi \in \mathsf{P}([r])} (|\Pi| - 1)! \prod_{A \in \Pi} |A|!\right) \prod_{j=1}^{r} \|B_j\|.$$

For partitions with k blocks,

$$\sum_{\substack{\Pi \in \mathsf{P}([r]) \\ |\Pi| = k}} \prod_{A \in \Pi} |A|! = \frac{r!}{k!} \binom{r-1}{k-1}.$$

Hence the combinatorial factor is

$$r! \sum_{k=1}^{r} \frac{1}{k} \binom{r-1}{k-1} = r! \frac{2^r - 1}{r} \le 2^r r!.$$

This proves (16). Repeated variables follow by polarization or by the same formula with repeated labels. □

For a multi-index α, call it even if

$$\sum_{I \ni a} \alpha_I \equiv 0 \pmod 2 \qquad \text{for all } a \in [N].$$

Only even multi-indices survive by Lemma 3.3.

Lemma 7.2 (Counting even marked hypergraphs at fixed order)**.** *Fix an even integer q and an integer $R \geq 1$, and set*

$$E_{q,R} = \left(\left\lfloor \frac{Rq}{2} \right\rfloor + 1 \right) \left(\frac{Rq}{2} \right)^{qR}.$$

Then, for every $1 \leq r \leq R$ and every N,

$$\sum_{\substack{|\alpha|=r \\ \alpha \text{ even}}} \frac{1}{\alpha!} \leq E_{q,R} N^{rq/2}.$$

Proof. If α is even and $|\alpha| = r$, the total marked incidence degree is rq and every vertex touched by the marked hyperedges has degree at least 2. Hence the union of the marked edges has size at most $rq/2$. The number of ordered r-tuples $(I_1, \dots, I_r)$ with even incidence is bounded by

$$\sum_{v \leq rq/2} \binom{N}{v} \binom{v}{q}^r \leq E_{q,R} N^{rq/2}, \qquad 1 \leq r \leq R.$$

A multi-index α with $|\alpha| = r$ corresponds to $r!/\alpha!$ ordered tuples. Therefore

$$\sum_{\substack{|\alpha|=r \\ \alpha \text{ even}}} \frac{1}{\alpha!} = \frac{1}{r!} \#\{(I_1, \dots, I_r) : \text{ the induced incidence is even}\},$$

and the preceding ordered-tuple bound proves the claim. □

Remark 7.3 (Why the fixed-order formulation is necessary)**.** *The estimate in Lemma 7.2 is intentionally stated with R fixed. A bound of the form $C_q^r N^{rq/2}/r!$, uniformly in r, is false in general. For instance, when $q = 4$, take disjoint two-point blocks $B_1, \dots, B_r$ and set $I_j = B_j \cup B_{j+1}$, with $B_{r+1} = B_1$. The marked incidence is even, and such cyclic marked cores already have entropy of order N^{2r}, with no compensating factor $1/r!$ available uniformly in r. This is why the finite-chaos argument below is valid only for fixed R. The present proof avoids any uniform all-orders marked-core counting by using the Hessian estimate coming from Assumption 1.3.*

Proposition 7.4 (Finite-chaos closure)**.** *Fix an integer $R \geq 3$ and a compact $K \subset \mathbb{R}$, and put $\beta_K = \sup_{\beta \in K} |\beta|$. Then*

$$\sup_{\beta \in K} \sum_{r=3}^{R} \|P_r L_N(\beta)\|_2^2 \leq E_{q,R} \sum_{r=3}^{R} (2^r r!)^2 \beta_K^{2r} ((q-1)!)^r N^{r(1-q/2)}. \tag{19}$$

In particular,

$$\sup_{\beta \in K} \sum_{r=3}^{R} \|P_r L_N(\beta)\|_2^2 = o(N^{2-q}). \tag{20}$$

Proof. For $|\alpha| = r$, write $\alpha = e_{I_1} + \cdots + e_{I_r}$ with repetitions. Applying Lemma 7.1 with $B_j = -\beta \sigma_N \psi_{I_j}$ gives, for $\beta \in K$ and $r \leq R$,

$$|\mathbb{E} \partial^\alpha L_N(\beta)| \leq 2^r r! \beta_K^r \sigma_N^r.$$

By Lemma 3.1 and orthogonality,

$$\|P_r L_N(\beta)\|_2^2 = \sum_{|\alpha|=r} \frac{1}{\alpha!} \left(\mathbb{E} \partial^\alpha L_N(\beta) \right)^2.$$

Lemma 3.3 restricts the sum to even α. Hence, using Lemma 7.2,

$$\|P_r L_N(\beta)\|_2^2 \leq E_{q,R} (2^r r!)^2 \beta_K^{2r} \sigma_N^{2r} N^{rq/2} = E_{q,R} (2^r r!)^2 \beta_K^{2r} ((q-1)!)^r N^{r(1-q/2)}. \tag{21}$$

Summing (21) gives (19). For $q \geq 4$ and $r \geq 3$, $r(1 - q/2) < 2 - q$, so the explicit bound implies (20). □

8 Proof attempt for the two-marked susceptibility estimate

This section pushes the proof of Assumption 1.3 to a precise high-temperature cluster-expansion input. The source-derivative identity, the finite-order Wick reductions, the Duhamel cumulant tree bound, and the connected parity counting lemma below are rigorous. These results explain why the desired estimate has the correct power of N and why the remaining small parameter should be β^2. They do not, however, by themselves prove the needed summability: repeated hyperedges make a coefficientwise or occurrence-level tree summation too crude. The only unproved input left in this section is therefore stated as a separate resummed two-replica L^2 cluster-expansion estimate, Assumption 8.17.

At $\beta = 0$, the Kubo covariance is diagonal. Indeed, Lemma 9.2 below gives

$$\mathcal{C}_{I,J}(0,g) = \operatorname{tr}(\psi_I \psi_J) = \mathbf{1}_{I=J}.$$

Hence

$$\sup_{I \in \mathcal{I}_N} \sum_{J \in \mathcal{I}_N} |\mathcal{C}_{I,J}(0,g)|^2 = 1.$$

The target estimate says that this bounded row-susceptibility persists, in annealed L^2, for sufficiently small $|\beta|$.

8.1 Source derivatives and finite-order connected coefficients

For fixed $I, J \in \mathcal{I}_N$ introduce source variables u, v and set

$$\Lambda_{I,J}(u,v;\beta,g) = \log \operatorname{tr} \exp\left(-\beta H_N(g) + u\psi_I + v\psi_J\right). \tag{22}$$

Lemma 8.1 (Kubo covariance as a source derivative)**.** *For every $I, J \in \mathcal{I}_N$,*

$$\mathcal{C}_{I,J}(\beta,g) = \partial_u \partial_v \Lambda_{I,J}(u,v;\beta,g)\big|_{u=v=0}.$$

Proof. First differentiate (22) with respect to u. By the Duhamel formula and cyclicity of trace,

$$\partial_u \Lambda_{I,J}(u,v;\beta,g)\big|_{u=v=0} = \langle \psi_I \rangle_{\beta,g}.$$

Differentiating this identity with respect to v and applying the Duhamel formula once more gives

$$\partial_u \partial_v \Lambda_{I,J}(u,v;\beta,g)\big|_{u=v=0} = \int_0^1 \left\{ \frac{\operatorname{tr}\left(\psi_I e^{-s\beta H_N(g)} \psi_J e^{-(1-s)\beta H_N(g)}\right)}{Z_N(\beta)} - \langle \psi_I \rangle_{\beta,g} \langle \psi_J \rangle_{\beta,g} \right\} ds.$$

This is exactly (1). □

For $m \geq 0$ and $K_1, \ldots, K_m \in \mathcal{I}_N$, define the connected Duhamel coefficient

$$\kappa(I,J;K_1,\ldots,K_m) = \partial_u \partial_v \partial_{t_1} \cdots \partial_{t_m} \log \operatorname{tr} \exp\left(u\psi_I + v\psi_J + \sum_{a=1}^m t_a \psi_{K_a} \right)\Bigg|_{u=v=t_1=\cdots=t_m=0}. \tag{23}$$

For $M \geq 0$, let the M-th formal truncation be

$$\mathcal{C}_{I,J}^{[M]}(\beta,g) = \sum_{m=0}^{M} \frac{(-\beta\sigma_N)^m}{m!} \sum_{K_1,\ldots,K_m \in \mathcal{I}_N} g_{K_1} \cdots g_{K_m} \kappa(I,J;K_1,\ldots,K_m). \tag{24}$$

Remark 8.2 (Why only truncations are asserted here)**.** *For fixed N and g, the source log-partition function is real analytic near the origin, but its complex Taylor radius need not be uniform in N or g. Therefore we do not claim, without an additional high-temperature expansion, that the infinite series obtained by sending $M \to \infty$ in (24) converges uniformly. The needed convergence is included explicitly in the resummed cluster-expansion input, Assumption 8.17, below.*

The word "connected" in (23) is literal. If the intersection hypergraph on the hyperedges $I, J, K_1, \dots, K_m$ splits into two vertex-disjoint subfamilies separating I from J, then

$$\kappa(I, J; K_1, \dots, K_m) = 0.$$

Indeed, normalized trace factorizes over disjoint Majorana tensor factors, and the logarithm extracts connected components by the exponential formula.

8.2 Cumulant structure of the Duhamel coefficients

We next isolate what can be proved directly about the coefficients (23). This is the useful replacement for an informal statement that "the logarithm keeps only connected diagrams." In particular, the sharp dependency-graph tree bound for the zero-source Duhamel coefficients is proved below.

Let $r \geq 1$ and let $E_1, \dots, E_r \in \mathcal{I}_N$. Put $A_a = \psi_{E_a}$. For $S \subset [r]$, define the symmetrized trace moment

$$\mu_{\mathbf{E}}(S) = \partial_{(t_a)_{a\in S}} \operatorname{tr} \exp\left(\sum_{a\in S} t_a A_a\right)\Big|_{t=0}. \tag{25}$$

Equivalently, if $S = \{a_1, \dots, a_s\}$, then

$$\mu_{\mathbf{E}}(S) = \frac{1}{s!} \sum_{\pi\in\mathfrak{S}_s} \operatorname{tr}\left(A_{a_{\pi(1)}} \cdots A_{a_{\pi(s)}}\right), \qquad \mu_{\mathbf{E}}(\emptyset) = 1. \tag{26}$$

Let $G_{\mathbf{E}}$ be the intersection graph on $[r]$, where $a \sim b$ iff $E_a \cap E_b \neq \emptyset$.

Lemma 8.3 (Duhamel coefficient as an ordinary cumulant)**.** *For $E_1, \dots, E_r \in \mathcal{I}_N$,*

$$\partial_{t_1} \cdots \partial_{t_r} \log \operatorname{tr} \exp\left(\sum_{a=1}^{r} t_a \psi_{E_a}\right)\Big|_{t=0} = \sum_{\pi\in\Pi_r} (|\pi| - 1)!(-1)^{|\pi|-1} \prod_{B\in\pi} \mu_{\mathbf{E}}(B), \tag{27}$$

where Π_r denotes the set of set partitions of $[r]$.

Proof. Set

$$M(t_1, \dots, t_r) = \operatorname{tr} \exp\left(\sum_{a=1}^{r} t_a \psi_{E_a}\right).$$

Since $M(0) = 1$, the Taylor coefficient of $\log M$ is obtained from the standard formal identity

$$\log M = \sum_{k\geq 1} \frac{(-1)^{k-1}}{k} (M - 1)^k. \tag{28}$$

Taking the coefficient of $t_1 \cdots t_r$ in (28) gives the partition formula (27). The moments that enter are precisely (25), because differentiating the exponential at the origin gives the symmetrized trace expression (26). □

Lemma 8.4 (Factorization and parity)**.** *The moments $\mu_{\mathbf{E}}$ satisfy the following two properties.*

First, if $S \subset [r]$ and $C_1, \dots, C_\ell$ are the connected components of $G_{\mathbf{E}}[S]$, then

$$\mu_{\mathbf{E}}(S) = \prod_{a=1}^{\ell} \mu_{\mathbf{E}}(C_a). \tag{29}$$

Second,

$$\mu_{\mathbf{E}}(S) = 0 \qquad \textit{unless} \qquad \triangle_{a\in S} E_a = \emptyset, \tag{30}$$

where $\triangle$ denotes symmetric difference. Consequently,

$$\partial_{t_1} \cdots \partial_{t_r} \log \operatorname{tr} \exp\left(\sum_{a=1}^{r} t_a \psi_{E_a}\right)\Big|_{t=0} = 0$$

whenever $G_{\mathbf{E}}$ is disconnected or $\triangle_{a=1}^{r} E_a \neq \emptyset$.

Proof. If $C_1, \dots, C_\ell$ are the connected components of $G_{\mathbf{E}}[S]$, then the Majorana labels used by different C_a are disjoint. The Clifford algebra representation decomposes as a tensor product over these disjoint label sets, and the normalized trace factorizes. Since monomials supported on different components commute, the symmetrized product in (26) factorizes as well. This proves (29).

For the parity statement, suppose some Majorana label b belongs to an odd number of sets E_a, $a \in S$. Flip the sign of ψ_b and leave all other Majorana generators fixed. This is a Clifford automorphism and hence is unitarily implemented. Therefore normalized traces are invariant under this operation. On the other hand every product appearing in (26) changes sign. Hence $\mu_{\mathbf{E}}(S) = -\mu_{\mathbf{E}}(S)$, proving (30).

Finally, if $G_{\mathbf{E}}$ is disconnected, then (29) says that the moment functional factorizes across a nontrivial partition of $[r]$. The cumulant formula (27) then vanishes by the usual cancellation of cumulants for a product moment functional. If $\triangle_{a=1}^{r} E_a \neq \emptyset$, then every block product in (27) has at least one block with odd total incidence, so it vanishes by (30). □

Lemma 8.5 (A crude absolute cumulant bound)**.** *For all $E_1, \dots, E_r \in \mathcal{I}_N$,*

$$\left| \partial_{t_1} \cdots \partial_{t_r} \log \operatorname{tr} \exp\left(\sum_{a=1}^{r} t_a \psi_{E_a} \right) \Big|_{t=0} \right| \leq e^r r!. \tag{31}$$

Proof. Since $\|\psi_E\| \leq 1$ for every $E \in \mathcal{I}_N$, each symmetrized trace moment satisfies $|\mu_{\mathbf{E}}(S)| \leq 1$. Hence (27) gives

$$|\kappa_r| \leq \sum_{\pi \in \Pi_r} (|\pi| - 1)!.$$

The number of partitions of $[r]$ into k blocks is at most $k^r/k!$. Therefore

$$\sum_{\pi \in \Pi_r} (|\pi| - 1)! \leq \sum_{k=1}^{r} \frac{k^r}{k!} (k-1)! \leq \sum_{k=1}^{r} k^{r-1} \leq r^r \leq e^r r!,$$

where the last inequality follows from Stirling's lower bound. This proves (31). □

The preceding lemmas are rigorous and show exactly what the connected Duhamel coefficient knows: it has a dependency graph, it vanishes off connected parity-even families, and it has at most factorial growth. We now prove the sharper tree-graph estimate that was previously left as a hypothesis.

Lemma 8.6 (Chromatic derivative bound)**.** *Let G be a finite graph. If G is connected, then*

$$|\chi_G'(0)| \leq \tau(G),$$

where χ_G is the chromatic polynomial and $\tau(G)$ is the number of spanning trees of G. If G is disconnected, then $\chi_G'(0) = 0$.

Proof. If G is disconnected, then $\chi_G(z)$ is divisible by z^2, and hence $\chi_G'(0) = 0$. Assume that G is connected. Fix an ordering of the edge set. Whitney's broken-circuit theorem writes the chromatic polynomial as

$$\chi_G(z) = \sum_{A} (-1)^{|A|} z^{c(A)},$$

where the sum is over edge subsets A containing no broken circuit and $c(A)$ is the number of connected components of the spanning subgraph $([r], A)$. The coefficient of z therefore counts, up to sign, the no-broken-circuit edge sets A for which $([r], A)$ is connected. Such an A is necessarily a spanning tree: if it were connected and contained a cycle, the largest edge in that cycle would produce a broken circuit. Hence $|\chi_G'(0)|$ is bounded by the number of spanning trees of G. □

Lemma 8.7 (Abstract dependency-graph cumulant tree bound)**.** *Let G be a graph on $[r]$. Let $\mu : 2^{[r]} \to \mathbb{C}$ satisfy $\mu(\emptyset) = 1$, $|\mu(S)| \le 1$ for all $S \subset [r]$, and*

$$\mu(S) = \prod_{C \in \mathrm{Comp}(G[S])} \mu(C) \tag{32}$$

for every $S \subset [r]$. Define

$$\kappa_\mu([r]) = \sum_{\pi \in \Pi_r} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{B \in \pi} \mu(B).$$

Then

$$|\kappa_\mu([r])| \le 2^{r-1}\tau(G),$$

where $\tau(G)$ is the number of spanning trees of G, and $\tau(G) = 0$ if G is disconnected.

Proof. For a partition π of $[r]$, let π^G be the partition obtained by replacing each block $B \in \pi$ by the connected components of $G[B]$. By (32),

$$\prod_{B \in \pi} \mu(B) = \prod_{C \in \pi^G} \mu(C).$$

Hence

$$\kappa_\mu([r]) = \sum_{\rho} a_G(\rho) \prod_{C \in \rho} \mu(C), \tag{33}$$

where the sum is over partitions ρ whose blocks are connected in G, and

$$a_G(\rho) = \sum_{\pi : \pi^G = \rho} (-1)^{|\pi|-1}(|\pi|-1)!.$$

Let $\overline{G}_\rho$ be the simple quotient graph whose vertices are the blocks of ρ, with an edge between two blocks if G has at least one edge between them. A partition π satisfying $\pi^G = \rho$ is the same as a partition of the vertices of $\overline{G}_\rho$ into nonempty independent sets. If $S_{\overline{G}_\rho}(k)$ denotes the number of such partitions into k independent sets, then

$$a_G(\rho) = \sum_{k \ge 1} (-1)^{k-1}(k-1)! S_{\overline{G}_\rho}(k).$$

The chromatic polynomial has the falling-factorial expansion

$$\chi_{\overline{G}_\rho}(z) = \sum_{k \ge 1} S_{\overline{G}_\rho}(k)(z)_k, \qquad (z)_k = z(z-1)\cdots(z-k+1).$$

Since $(z)_k'|_{z=0} = (-1)^{k-1}(k-1)!$, we get

$$a_G(\rho) = \chi'_{\overline{G}_\rho}(0).$$

By Lemma 8.6,

$$|a_G(\rho)| = |\chi'_{\overline{G}_\rho}(0)| \le \tau(\overline{G}_\rho), \tag{34}$$

with the convention that $\tau(\overline{G}_\rho) = 0$ when the quotient graph is disconnected.

Since $|\mu(C)| \le 1$, (33) and (34) give

$$|\kappa_\mu([r])| \le \sum_{\rho} \tau(\overline{G}_\rho), \tag{35}$$

where only ρ with connected quotient contribute. It remains to bound this sum by $2^{r-1}\tau(G)$.

For each connected subset $C \subset [r]$, fix once and for all a spanning tree T_C of $G[C]$. For each pair of disjoint subsets C, D with at least one edge between them, fix one canonical edge of G joining C to D. Given a partition ρ with connected blocks and a spanning tree $\overline{T}$ of $\overline{G}_\rho$, form a spanning tree of G by taking the union of all T_C, $C \in \rho$, and the canonical inter-block edges corresponding to the edges of $\overline{T}$. Mark the inter-block edges in this tree. This maps each pair $(\rho, \overline{T})$ injectively into a pair (T, F), where T is a spanning tree of G and $F \subset E(T)$. Indeed, removing F from T recovers the blocks of ρ, and F recovers $\overline{T}$. Therefore

$$\sum_{\rho} \tau(\overline{G}_\rho) \le \#\{(T, F) : T \in \mathsf{Tree}(G),\ F \subset E(T)\} = 2^{r-1}\tau(G). \tag{36}$$

Combining (35) and (36) proves the lemma. □

Proposition 8.8 (Duhamel cumulant tree-graph inequality)**.** *For every $r \ge 2$ and every $E_1, \dots, E_r \in \mathcal{I}_N$,*

$$\left| \partial_{t_1} \cdots \partial_{t_r} \log \operatorname{tr} \exp\left(\sum_{a=1}^{r} t_a \psi_{E_a} \right) \Big|_{t=0} \right| \le 2^{r-1} \sum_{T \in \mathsf{Tree}([r])} \prod_{\{a,b\} \in E(T)} \mathbf{1}_{E_a \cap E_b \neq \emptyset}. \tag{37}$$

Proof. Let $G = G_{\mathbf{E}}$ be the intersection graph. Since $\|\psi_E\| \le 1$, the symmetrized trace moments satisfy $|\mu_{\mathbf{E}}(S)| \le 1$. By Lemma 8.4, they factorize over connected components of $G[S]$. Lemma 8.3 identifies the left-hand side of (37) with the ordinary cumulant of this moment functional. Lemma 8.7 therefore gives

$$|\kappa_D(E_1, \dots, E_r)| \le 2^{r-1}\tau(G).$$

Finally,

$$\tau(G) = \sum_{T \in \mathsf{Tree}([r])} \prod_{\{a,b\} \in E(T)} \mathbf{1}_{E_a \cap E_b \neq \emptyset}.$$

Thus (37) holds. □

Remark 8.9 (What this proves and what it does not prove)**.** *Proposition 8.8 closes the single-replica Duhamel tree bound. It does not, by itself, close the two-replica susceptibility. After Wick pairing, one has two Duhamel trees and a pairing of their unmarked occurrences. A naive summation over occurrence-level tree pairs loses a superexponential factor because repeated hyperedges are counted too many times. The connected parity count below gives the correct power of N, but the remaining summability must be proved by the resummed two-replica cluster-expansion estimate isolated in Assumption 8.17.*

8.3 The finite two-replica Wick reduction

Squaring (24) and taking Gaussian expectation produces two copies of the connected expansion. The Gaussian expectation is computed by Wick pairing. It is useful to package the result as diagrams.

A two-replica diagram Γ with marks I, J consists of two ordered lists of unmarked hyperedges,

$$\mathbf{K} = (K_1, \dots, K_m), \qquad \mathbf{L} = (L_1, \dots, L_{m'}),$$

paired by a Wick pairing of the multiset $\{K_1, \dots, K_m, L_1, \dots, L_{m'}\}$. In particular $m + m' = 2p$ is even. Its value is

$$W(\Gamma) = \frac{\kappa(I, J; K_1, \dots, K_m)\kappa(I, J; L_1, \dots, L_{m'})}{m!m'!}. \tag{38}$$

Let $\mathfrak{D}_p^{[M]}(I, J)$ denote the set of such diagrams with $m, m' \le M$ and with p Wick pairs for which both connected Duhamel coefficients in (38) are nonzero.

Lemma 8.10 (Finite annealed two-replica formula)**.** *For fixed M, N, I, J and $\beta \in \mathbb{R}$,*

$$\mathbb{E}\left[\left|\mathcal{C}_{I,J}^{[M]}(\beta, g)\right|^2\right] = \sum_{p=0}^{M} (\beta\sigma_N)^{2p} \sum_{\Gamma \in \mathfrak{D}_p^{[M]}(I,J)} W(\Gamma). \tag{39}$$

Consequently,

$$\mathbb{E}\left[\left|\mathcal{C}_{I,J}^{[M]}(\beta, g)\right|^2\right] \le \sum_{p=0}^{M} |\beta|^{2p} \sigma_N^{2p} \sum_{\Gamma \in \mathfrak{D}_p^{[M]}(I,J)} |W(\Gamma)|.$$

Proof. The coefficients $\kappa(I, J; K_1, \dots, K_m)$ are real. Indeed, they are Taylor coefficients at the origin of the real analytic function

$$(u, v, \lambda_1, \dots, \lambda_m) \mapsto \log \operatorname{tr} \exp\left(u\psi_I + v\psi_J + \sum_{a=1}^{m} \lambda_a \psi_{K_a}\right)$$

restricted to real sources. Thus the square on the left-hand side is the ordinary product of the two real finite expansions.

Insert (24) twice. For two ordered lists $\mathbf{K}$ and $\mathbf{L}$, the Gaussian expectation

$$\mathbb{E}\left[g_{K_1} \cdots g_{K_m} g_{L_1} \cdots g_{L_{m'}}\right] \tag{40}$$

is zero unless $m+m'$ is even. If $m+m' = 2p$, Wick's theorem writes (40) as the sum over all pairings of the multiset of labels. Grouping each Wick pairing with the two ordered lists gives $\mathfrak{D}_p^{[M]}(I, J)$ and the value (38). This proves (39). The absolute bound follows by the triangle inequality. □

8.4 Connected parity counting

The next lemma is the purely combinatorial count that explains why the parity constraint determining J removes exactly one extra hyperedge degree of freedom. It is useful in any attempt to prove the cluster-expansion estimate below.

Lemma 8.11 (Connected parity counting)**.** *Fix $I \in \mathcal{I}_N$, $p \ge 0$, and $C \subset [p]$. For $H_1, \dots, H_p \in \mathcal{I}_N$, set*

$$J = I \triangle \triangle_{a \in C} H_a.$$

Let $N_p(I, C)$ be the number of ordered p-tuples $(H_1, \dots, H_p)$ such that $J \in \mathcal{I}_N$ and the intersection graph on the labelled hyperedges

$$I, \qquad J, \qquad H_1, \dots, H_p$$

is connected. Then

$$N_p(I, C) \le (2e^2 q^2)^p p! N^{(q-1)p}.$$

Proof. The case $p = 0$ is immediate. Assume $p \ge 1$ and consider a tuple counted by $N_p(I, C)$. We first prove

$$\left|\left(\bigcup_{a=1}^{p} H_a\right) \setminus I\right| \le (q-1)p. \tag{41}$$

Consider the intersection graph on the labelled hyperedges $I, H_1, \dots, H_p$, without J, and let its connected components be $\mathcal{D}_1, \dots, \mathcal{D}_\ell$. A hyperedge in such a component is called selected if it is either I or H_a with $a \in C$. For a component $\mathcal{D}$, let $e(\mathcal{D})$ be the number of its hyperedges and let $c(\mathcal{D})$ be the number of its selected hyperedges. Define

$$R(\mathcal{D}) = \triangle_{E \in \mathcal{D},\ E \text{ selected}} E.$$

Since distinct components have disjoint supports, the sets $R(\mathcal{D}_j)$ are disjoint and

$$J = \bigsqcup_{j=1}^{\ell} R(\mathcal{D}_j).$$

Hence

$$\sum_{j=1}^{\ell} |R(\mathcal{D}_j)| = |J| = q. \tag{42}$$

Each component $\mathcal{D}_j$ contains at least one selected hyperedge. If it did not, then $R(\mathcal{D}_j) = \emptyset$, and the support of $\mathcal{D}_j$ would be disjoint from J and from all other components. This would contradict the connectedness of the intersection graph on $I, J, H_1, \dots, H_p$. Thus $c(\mathcal{D}_j) \geq 1$ for all j.

Fix a component $\mathcal{D}$. Let $U_{\mathrm{sel}}(\mathcal{D})$ be the union of its selected hyperedges. Since the total selected incidence is $qc(\mathcal{D})$, every point in $R(\mathcal{D})$ contributes at least one incidence and every point in $U_{\mathrm{sel}}(\mathcal{D}) \setminus R(\mathcal{D})$ contributes at least two, so

$$|U_{\mathrm{sel}}(\mathcal{D})| \leq \frac{qc(\mathcal{D}) + |R(\mathcal{D})|}{2}.$$

The non-selected hyperedges in $\mathcal{D}$ can be ordered so that each new one intersects the union of the previously exposed selected and non-selected hyperedges. Each such hyperedge adds at most $q-1$ new vertices. Therefore

$$\begin{aligned}\left|\bigcup_{E\in\mathcal{D}} E\right| &\leq \frac{qc(\mathcal{D}) + |R(\mathcal{D})|}{2} + (q-1)(e(\mathcal{D}) - c(\mathcal{D})) \\ &= (q-1)e(\mathcal{D}) - \frac{q-2}{2}c(\mathcal{D}) + \frac{1}{2}|R(\mathcal{D})|.\end{aligned}$$

Summing over all components and using $\sum_j e(\mathcal{D}_j) = p+1$, $\sum_j c(\mathcal{D}_j) = |C|+1$, and (42), we obtain

$$\begin{aligned}\left|I \cup \bigcup_{a=1}^{p} H_a\right| &\leq (q-1)(p+1) - \frac{q-2}{2}(|C|+1) + \frac{q}{2} \\ &\leq q + (q-1)p.\end{aligned}$$

Since $|I| = q$, this proves (41).

We now count. Encode $(H_1, \dots, H_p)$ by writing each H_a in increasing order, giving pq positions. By (41), at most $(q-1)p$ distinct labels outside I appear. Choose one representative position for each such outside label and, if necessary, add arbitrary positions until exactly $(q-1)p$ representative positions have been chosen. The representative positions can be chosen in at most

$$\binom{pq}{(q-1)p} = \binom{pq}{p} \leq (eq)^p$$

ways. Their labels can be chosen in at most $N^{(q-1)p}$ ways. Each of the remaining p positions is either one of the q labels in I or copies one of the representative outside labels, hence has at most $q + (q-1)p \leq qp + q$ choices. Thus

$$N_p(I, C) \leq (eq)^p (qp+q)^p N^{(q-1)p}.$$

Since $p + 1 \leq 2p$ and $p^p \leq e^p p!$ for $p \geq 1$, this is bounded by $(2e^2q^2)^p p! N^{(q-1)p}$. □

Remark 8.12 (Use and limitation of the parity count)**.** *Lemma 8.11 gives the correct N-power for Wick-paired diagrams: the cross-pair set C determines $J = I \triangle \triangle_{a\in C} H_a$, so summing over J costs no extra factor N^{q-1}. This is exactly the power needed to cancel $\sigma_N^{2p} = ((q-1)!)^p N^{-(q-1)p}$. However, the lemma counts only label assignments. It does not by itself control the entropy of two occurrence-level Duhamel trees after Wick pairing. That remaining entropy is exactly why the remaining input is stated as a resummed cluster-expansion estimate below.*

The finite two-replica formula is exact for every finite truncation. It is also useful as a diagnostic for what remains unproved. If one bounds the two zero-source Duhamel cumulants by occurrence-level trees and then sums the trees absolutely after Wick pairing, repeated hyperedges create a superexponential entropy loss. For example, a configuration in which all $2p$ unmarked occurrences are the same hyperedge has an occurrence-level complete graph and therefore about $(2p)^{2p}$ spanning trees, whereas the Duhamel and Wick factorials do not reduce this to a purely exponential factor. This does not show that the two-marked susceptibility is false; it shows that the coefficientwise tree-counting strategy is too crude. The remaining input must therefore be a resummed two-replica L^2 cluster expansion, where repeated hyperedges are treated before taking absolute values of the final cluster weights.

8.5 Why the remaining estimate is not proved by the naive expansions

The preceding lemmas prove the two structural facts one expects from a high-temperature expansion: connectedness and the correct parity count. It is natural to try to finish the proof by taking absolute values term by term in a single-replica expansion, or by applying the Duhamel tree bound coefficient by coefficient. The following two observations explain why these two routes do not close. They do not disprove the resummed cluster expansion; rather, they show why the last input must be a genuinely annealed two-replica estimate.

Lemma 8.13 (Single-replica absolute activities are too large)**.** *Let*

$$\eta_K(g) = \exp\left(c|\beta|\sigma_N|g_K|\right) - 1$$

for a fixed constant $c > 0$. For every fixed $I \in \mathcal{I}_N$ and every fixed $\beta \neq 0$,

$$\sum_{K \in \mathcal{I}_N : K \cap I \neq \emptyset} \mathbb{E}\eta_K(g) \geq \frac{cq|\beta|}{2} \sqrt{\frac{2}{\pi(q-1)!}}\, N^{(q-1)/2} \tag{43}$$

for all large N.

Proof. Since $e^x - 1 \geq x$ for $x \geq 0$ and $\mathbb{E}|g_K| = \sqrt{2/\pi}$,

$$\mathbb{E}\eta_K(g) \geq c|\beta|\sigma_N \sqrt{\frac{2}{\pi}}.$$

On the other hand,

$$\#\{K \in \mathcal{I}_N : K \cap I \neq \emptyset\} = \binom{N}{q} - \binom{N-q}{q} \geq \frac{q}{2(q-1)!} N^{q-1}$$

for all large N, because the left-hand side is a polynomial in N with leading coefficient $q/(q-1)!$. Using $\sigma_N = ((q-1)!)^{1/2} N^{-(q-1)/2}$ gives (43). □

Remark 8.14. *Lemma 8.13 rules out a standard deterministic Kotecký–Preiss proof based on absolute single-edge activities. The small parameter cannot be σ_N itself. It must be the Wick-paired parameter $\sigma_N^2 N^{q-1} = O_q(1)$ together with β^2.*

The next observation records the other failure mode. Even after identifying connected Duhamel cumulants, expanding repeated hyperedges coefficient by coefficient is too fine a decomposition.

Lemma 8.15 (Coefficientwise distinct-support tree bounds are too strong)**.** *Fix even $q \geq 4$. There is no constant $A_q < \infty$ with the following property. For every family of distinct q-sets $E_0, E_1, \dots, E_s$ and every multi-index $a = (a_1, \dots, a_s)$, the coefficient*

$$[z_1^{a_1} \cdots z_s^{a_s}] \partial_u \partial_v \log \operatorname{tr} \exp\left((u+v)\psi_{E_0} + \sum_{j=1}^{s} z_j \psi_{E_j} \right) \Big|_{u=v=0}$$

is bounded in absolute value by

$$A_q^{|a|}\,\tau(G), \tag{44}$$

where G is the intersection graph on $\{E_0, E_1, \dots, E_s\}$ and $\tau(G)$ is its number of spanning trees.

Proof. Choose a fixed set $S \subset [N]$ with $|S| = q-1$ and distinct points $x_0, x_1, \dots, x_s \notin S$. Set

$$E_j = S \cup \{x_j\}, \qquad A_j = \psi_{E_j}.$$

Since $|E_i \cap E_j| = q-1$ is odd for $i \neq j$, the operators $A_0, A_1, \dots, A_s$ anticommute pairwise. Also $A_j^2 = \mathbb{I}$. Therefore

$$\operatorname{tr}\exp\left(x_0 A_0 + \sum_{j=1}^{s} z_j A_j\right) = \cosh\left(\sqrt{x_0^2 + \sum_{j=1}^{s} z_j^2}\right). \tag{45}$$

Writing

$$h_q(w) = \log\cosh\left(\sqrt{w}\right),$$

we get

$$\partial_u \partial_v \log\operatorname{tr}\exp\left((u+v)A_0 + \sum_{j=1}^{s} z_j A_j\right)\Big|_{u=v=0} = 2h_q'\left(\sum_{j=1}^{s} z_j^2\right). \tag{46}$$

The function h_q' has its nearest singularity at a negative real point, coming from the first zero of cosh. Hence, by elementary singularity analysis, there are constants $b_q, c_q > 0$ such that the coefficient b_m of w^m in $h_q'(w)$ satisfies

$$|b_m| \geq b_q c_q^m$$

for all sufficiently large m. Take $a_1 = \cdots = a_s = 2k$ and $m = ks$. The coefficient of $z_1^{2k} \cdots z_s^{2k}$ in (46) is

$$2b_{ks}\frac{(ks)!}{(k!)^s}.$$

By Stirling's formula this is at least

$$(c_q')^{ks} s^{ks} \tag{47}$$

up to subexponential factors in k and s. In the present example the intersection graph is the complete graph on $s+1$ vertices, so

$$\tau(G) = (s+1)^{s-1}.$$

For any fixed A_q, choose s large enough and then k large enough. The lower bound (47) is then larger than $A_q^{2ks}(s+1)^{s-1}$, contradicting (44). □

Remark 8.16. *Lemma 8.15 is the precise reason that the final estimate below is stated after resummation. The example is harmless for the actual covariance: it is controlled by the closed form (45). It only becomes dangerous if one first expands the repeated supports into individual Taylor coefficients and then takes absolute values.*

Assumption 8.17 (Resummed two-replica L^2 cluster expansion)**.** *This is the unproved high-temperature input in the present CLT proof. There exist constants $B_q < \infty$ and $\beta_{\mathrm{ce}} > 0$, depending only on q, such that the following statements hold for $|\beta| < \beta_{\mathrm{ce}}$.*

First, for all N and $I, J \in \mathcal{I}_N$,

$$\mathcal{C}_{I,J}^{[M]}(\beta, g) \to \mathcal{C}_{I,J}(\beta, g) \qquad \text{in } L^2(\,d\gamma(g)) \text{ as } M \to \infty. \tag{48}$$

Second, for each $M \geq 0$ there is a decomposition in $L^2(\,d\gamma(g))$ of the finite high-temperature truncation into resummed two-marked cluster layers,

$$\mathcal{C}_{I,J}^{[M]}(\beta, g) = \sum_{p=0}^{M} \mathcal{A}_{p;I,J}^{[M]}(\beta, g),$$

where $\mathcal{A}^{[M]}_{p;I,J}$ is the contribution of Wick-paired connected parity clusters with p Gaussian pairs after all repeated-hyperedge multiplicities inside each connected cluster have been resummed. More formally, the decomposition is required to be compatible with the exact finite Wick formula of Lemma 8.10: the p-th layer is obtained by collecting all terms with p Wick pairs and by summing, before taking absolute values, all diagrams which have the same connected support and differ only by repeated occurrences of hyperedges inside that support. Thus $\mathcal{A}^{[M]}_{0;I,J}$ is the zero-disorder covariance layer, supported on the diagonal mark $I = J$, while $\mathcal{A}^{[M]}_{1;I,J}$ is the first annealed correction with one Gaussian pair. These examples are meant only to fix the interpretation of the layers; the estimate below is the actual hypothesis.

The layers satisfy the row-wise L^2 estimate

$$\sup_{M\geq 0}\sup_{N\geq 1}\sup_{I\in\mathcal{I}_N}\sum_{J\in\mathcal{I}_N}\mathbb{E}\left[\left|\mathcal{A}^{[M]}_{p;I,J}(\beta,g)\right|^2\right]\leq B_q^p|\beta|^{2p}\qquad \text{for every } p\geq 0. \tag{49}$$

Remark 8.18 (Meaning of the cluster-expansion assumption)**.** *Assumption 8.17 is not a coefficientwise tree bound. Such a bound is too strong, because repeated hyperedges can make individual Taylor coefficients very large before they are resummed. The assumption asks for the natural object instead: an L^2 estimate on the resummed two-replica cluster layers. This is plausible for small $|\beta|$ because each Wick pair contributes the factor*

$$\sigma_N^2=(q-1)!N^{-(q-1)},$$

while a new connected q-hyperedge has at most order N^{q-1} choices once it is required to touch the already exposed cluster. Thus the N-powers cancel pair by pair and the remaining expansion should be subcritical for small $|\beta|$. This is the two-marked L^2 analogue of the sparse connected-component summability used in the high-temperature free-energy expansion of [GPSZ26]. The commutation-index estimates of [ACKK25] are not strong enough by themselves to imply (49); they control Lipschitz concentration, whereas (49) is a sharper annealed two-point susceptibility estimate.

Proposition 8.19 (Cluster expansion implies two-marked susceptibility)**.** *Assumption 8.17 implies Assumption 1.3, with any*

$$\beta_{\mathrm{sus}}<\min\{\beta_{\mathrm{ce}},B_q^{-1/2}\}. \tag{50}$$

More precisely, if $K\subset(-\beta_{\mathrm{sus}},\beta_{\mathrm{sus}})$ is compact and $\beta_K=\sup_{\beta\in K}|\beta|$, then the susceptibility constant in (2) satisfies

$$S_{q,K}\leq\left(1-B_q^{1/2}\beta_K\right)^{-2}. \tag{51}$$

Proof. Fix $|\beta|<\min\{\beta_{\mathrm{ce}},B_q^{-1/2}\}$. For the finite truncation, Minkowski's inequality in the Hilbert space $L^2(\,d\gamma(g);\ell^2(\mathcal{I}_N))$ gives, uniformly in M,N,I,

$$\begin{aligned}\left(\sum_{J\in\mathcal{I}_N}\mathbb{E}\left[\left|\mathcal{C}^{[M]}_{I,J}(\beta,g)\right|^2\right]\right)^{1/2}&\leq\sum_{p=0}^{M}\left(\sum_{J\in\mathcal{I}_N}\mathbb{E}\left[\left|\mathcal{A}^{[M]}_{p;I,J}(\beta,g)\right|^2\right]\right)^{1/2}\\&\leq\sum_{p=0}^{M}(B_q^{1/2}|\beta|)^p.\end{aligned}$$

The right-hand side is bounded uniformly in M,N,I. Taking $M\to\infty$ and using (48), we obtain

$$\sum_{J\in\mathcal{I}_N}\mathbb{E}\left[|\mathcal{C}_{I,J}(\beta,g)|^2\right]\leq\left(\sum_{p\geq 0}(B_q^{1/2}|\beta|)^p\right)^2.$$

Taking the supremum over $\beta\in K\subset(-\beta_{\mathrm{sus}},\beta_{\mathrm{sus}})$, with β_{sus} satisfying (50), gives (51) and proves Assumption 1.3. □

Remark 8.20 (What is proved and what remains). *The exact source-derivative representation, the finite connected Duhamel coefficients, the dependency-graph Duhamel tree bound, the connected parity counting lemma, and the finite Wick-paired two-replica formula are proved above. These results justify the structure and the expected N-scaling of the remaining cluster expansion. What is still not proved is Assumption 8.17: the construction of a resummed two-replica expansion for the Kubo covariance and the uniform row-wise L^2 estimate (49). This is the genuinely new high-temperature input needed to make the CLT unconditional.*

9 Small-β consistency check

Lemma 9.1 (Trace of a nontrivial Majorana monomial). *If $A \subset [N]$ is nonempty, then*

$$\operatorname{tr}\left(\prod_{a\in A}\psi_a\right) = 0.$$

Proof. Choose $a \in A$. By the sign-flip unitary used in Lemma 3.2, there is a unitary U such that $U\psi_a U^* = -\psi_a$ and $U\psi_b U^* = \psi_b$ for $b \neq a$. Hence the monomial changes sign under conjugation by U. Trace invariance gives the claim. □

Lemma 9.2 (Trace orthogonality). *For $I, J \in \mathcal{I}_N$,*

$$\operatorname{tr}(\psi_I\psi_J) = \mathbf{1}_{I=J}.$$

Proof. If $I = J$, then $\psi_I^2 = \mathbb{I}$. If $I \neq J$, the product $\psi_I\psi_J$ is, up to a scalar, a nontrivial Majorana monomial supported on $I\triangle J$, so Lemma 9.1 applies. □

Proposition 9.3 (Leading small-β fluctuation). *As $\beta \to 0$ with N and g fixed,*

$$L_N(\beta) = \frac{\beta^2}{2}\operatorname{tr}(H_N^2) + \mathcal{R}_N(\beta), \qquad |\mathcal{R}_N(\beta)| \le 8|\beta|^3\,\|H_N\|^3\,, \tag{52}$$

and

$$\operatorname{tr}(H_N^2) = \sigma_N^2 \sum_{I\in\mathcal{I}_N} g_I^2. \tag{53}$$

Consequently the leading random term in $L_N(\beta) - \mathbb{E}L_N(\beta)$ is

$$\frac{\beta^2}{2}\sigma_N^2 \sum_{I\in\mathcal{I}_N}(g_I^2 - 1), \tag{54}$$

and

$$\operatorname{Var}\left[N^{q/2-1}\frac{\beta^2}{2}\sigma_N^2\sum_{I\in\mathcal{I}_N}(g_I^2-1)\right] \to \frac{((q-1)!)^2}{2q!}\beta^4. \tag{55}$$

Proof. Let $F(\beta) = \log\operatorname{tr} e^{-\beta H_N}$. Since $\operatorname{tr}(H_N) = 0$ by Lemma 9.1, we have $F(0) = F'(0) = 0$ and $F''(0) = \operatorname{tr}(H_N^2)$. Lemma 7.1, applied with $B_1 = B_2 = B_3 = H_N$, gives $|F'''(t)| \le 2^3 3!\,\|H_N\|^3$ for all real t. Taylor's formula gives the remainder bound in (52). By Lemma 9.2,

$$\operatorname{tr}(H_N^2) = \sigma_N^2 \sum_{I,J\in\mathcal{I}_N} g_I g_J \operatorname{tr}(\psi_I\psi_J) = \sigma_N^2 \sum_{I\in\mathcal{I}_N} g_I^2.$$

This proves (53) and (54). Since $\operatorname{Var}(g_I^2 - 1) = 2$,

$$\operatorname{Var}\left[N^{q/2-1}\frac{\beta^2}{2}\sigma_N^2\sum_I(g_I^2-1)\right] = N^{q-2}\frac{\beta^4}{4}\sigma_N^4 2M_N.$$

Using $\sigma_N^2 = (q-1)!N^{-(q-1)}$ and $M_N \sim N^q/q!$ gives (55). □

Remark 9.4 (Check against the covariance formula)**.** *The leading small-β free-energy density is*

$$\ell_q(\beta) = \frac{\beta^2}{2q} + O(\beta^4), \qquad \ell_q'(\beta) = \frac{\beta}{q} + O(\beta^3).$$

Hence

$$\frac{q!}{2}(\beta\ell_q'(\beta))^2 = \frac{((q-1)!)^2}{2q!}\beta^4 + O(\beta^6),$$

which matches (55)*.*

10 Conclusion

The proof of Theorem 1.4 is complete from the locally uniform LLN and the two-marked susceptibility estimate. The locally uniform LLN is the high-temperature free-energy input from the existing literature, and its β-derivative follows from convexity. The second chaos is exact and gives the rank-one Gaussian limit. Every fixed higher chaos is negligible by the parity count and the Duhamel derivative bound. The infinite chaos tail is controlled by the Hilbert–Schmidt norm of the Hessian through the Malliavin identity. Section 8 proves the single-replica Duhamel cumulant tree bound and the connected parity counting lemma needed for the two-marked susceptibility. The only remaining unclosed point is the resummed two-replica L^2 cluster-expansion estimate in Assumption 8.17, which also includes convergence of the high-temperature truncations to the actual Kubo covariance. Subsection 8.5 records the two main failed coefficientwise approaches, so the remaining assumption should be read as a genuine resummed Wick-pair polymer estimate rather than as a hidden elementary counting lemma.

References

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